

CENG 222

Statistical Methods for Computer Engineering

Week 12

Chapter 11 Regression

11.1 Least squares estimation

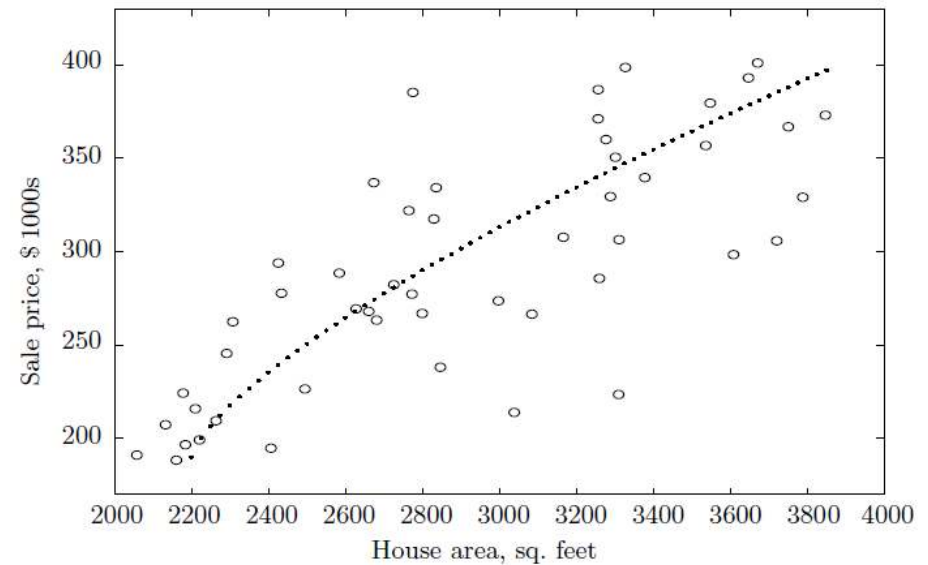
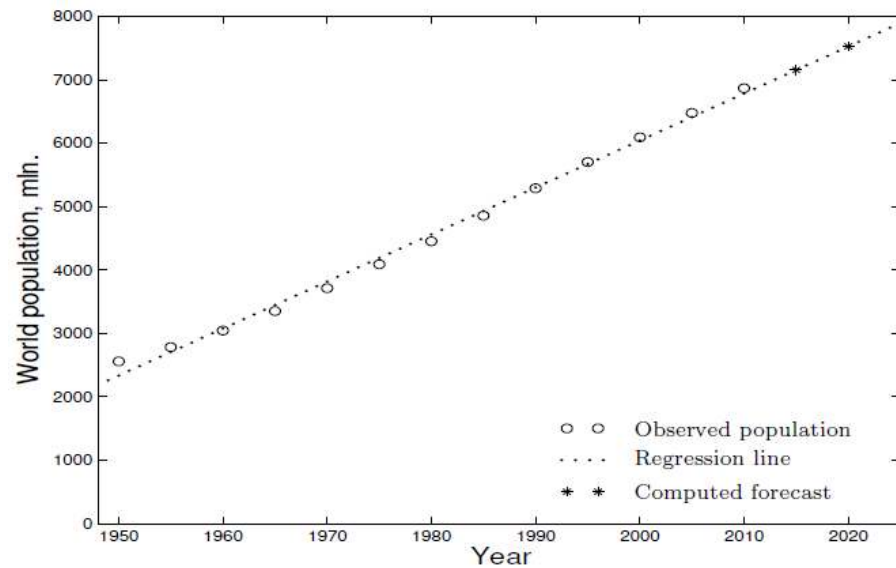
Regression

- Analysis of relations between random variables
- Regression of Y on $X^{(1)}, \dots, X^{(k)}$ is the conditional expectation:
 - $\mathbf{E}(Y|X^{(1)} = x^{(1)}, \dots, X^{(k)} = x^{(k)})$
 - Y is called the *response* or *dependent* variable. It is the variable we want to predict
 - $X^{(i)}$ s are called the *predictors* or *independent* variables.
- Linear multi-variate regression
 - $Y = \beta_0 + \beta_1 X^{(1)} + \beta_2 X^{(2)} + \dots + \beta_k X^{(k)}$

Regression

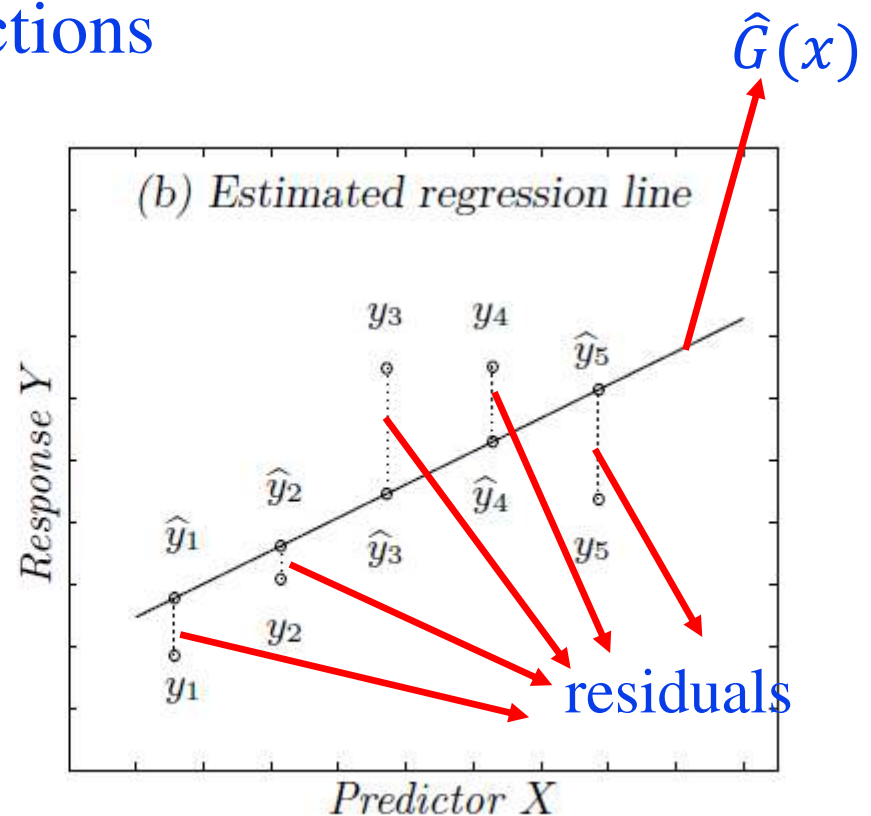
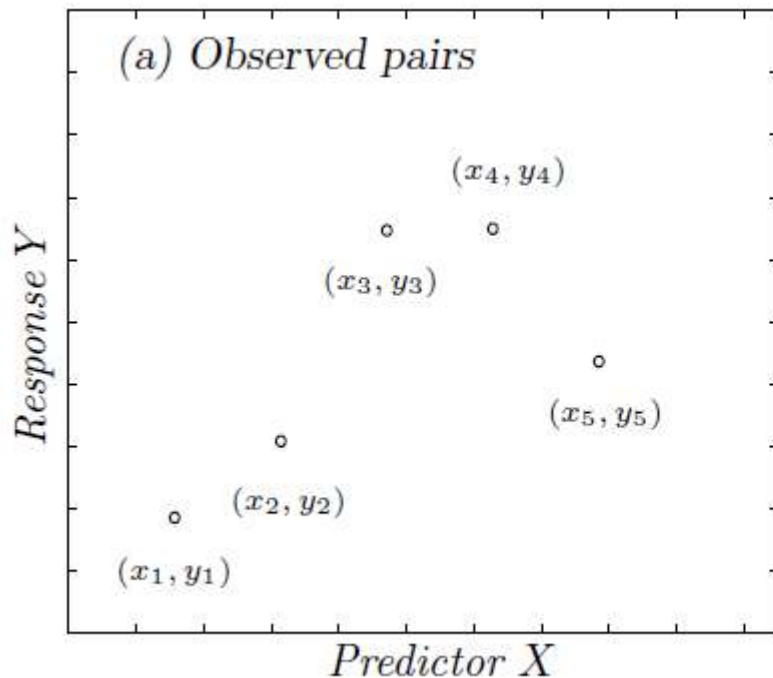
- In this course, we will cover the simplest form:
 - Univariate, linear regression
 - $G(x) = \mathbf{E}(Y|X = x) = \beta_0 + \beta_1 X$
 - Intercept: $\beta_0 = G(0)$
 - Slope: $\beta_1 = G(x + 1) - G(x)$

Linear versus Non-Linear Regression



Method of least squares

- Estimate the function $G(x)$ with $\hat{G}(x)$
 - $\hat{G}(x)$: try to minimize the distance between real observations and predictions



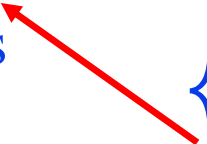
Method of least squares

- Find $\hat{G}(x)$ that minimizes the sum of squares of the residuals
 - $e_i = y_i - \hat{y}_i$
 - Minimize $\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

Method of least squares

- $Q = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{G}(x_i))^2$
 $= \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$
- Take partial derivatives wrt β_0 and β_1 and equate to 0

normal equations


$$\left\{ \begin{array}{l} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \end{array} \right.$$

Method of least squares

- $\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$

$$\rightarrow \beta_0 = \frac{\sum y_i - \beta_1 \sum x_i}{n} = \bar{y} - \beta_1 \bar{x}$$

- Substitute β_0 in the second normal equation:

$$\rightarrow \sum_{i=1}^n x_i ((y_i - \bar{y}) - \beta_1 (x_i - \bar{x})) = 0$$

$$\rightarrow S_{xy} - \beta_1 S_{xx} = 0 \text{ where}$$

sum of squares $\leftarrow S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$

sum of cross products $\leftarrow S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

Method of least squares steps

1. Compute $\bar{x}$ and $\bar{y}$
 2. $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$
 3. $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
 4. $b_1 = \hat{\beta}_1 = S_{xy}/S_{xx}$
 5. $b_0 = \hat{\beta}_0 = \bar{y} - b_1\bar{x}$
- Example 11.3 (World Population)

Regression and correlation

- Recall that covariance and correlation coefficient are:
 - $Cov(X, Y) = E \left((X - E(X))(Y - E(Y)) \right)$
 - $\rho = \frac{Cov(X, Y)}{\sigma_x \sigma_y}$
- Sample covariance and sample correlation coefficient can be used to estimate $Cov(X, Y)$ and ρ

Sample covariance and correlation coefficient

- $s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$
- $r = \frac{s_{xy}}{s_x s_y}$
- s_x and s_y are sample standard deviations
- $s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$ and $s_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}$
- $\rightarrow b_1 = \frac{s_{xy}}{s_{xx}} = \frac{s_{xy}}{s_x s_x} = r \left(\frac{s_y}{s_x} \right)$

Why linear when we can have 0 sum of errors?

- Answer: to avoid overfitting

